

Securing the Future: Building Antifragile Supply Chains in a Volatile World



Presented by:

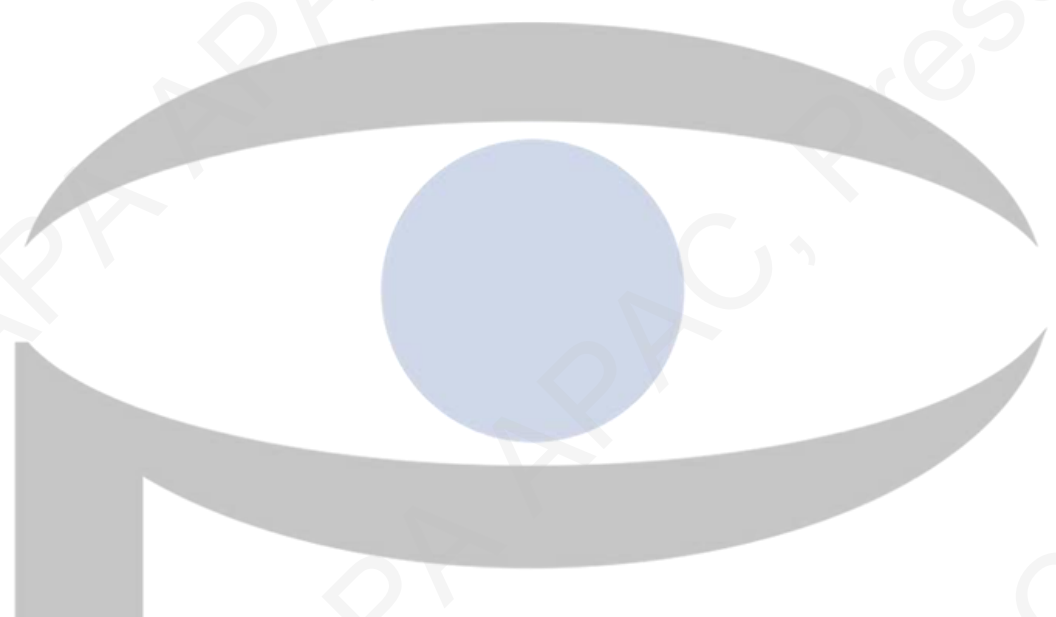
Alvin Lau

Executive Director, TAPA APAC



About Us

The Transported Asset Protection Association



- 850+**
TAPA APAC Members
- 1000+**
TAPA APAC Entities
- 2,200+**
TSR Certified Trucks
- 850+**
FSR Certified Sites
- 2,000+**
Trained TAPA Standards Authorized Auditors

ABOUT TAPA

The Leading Global Association For Supply Chain Resilience & Sustainability



- **Structured Audit Process**
- **Recognised by Global Government Agencies**
- **Independent Audit Bodies such as BV, SGS, Intertek, TUV**
- **LMS System**
- **TAPA Intelligence System**

Set up by manufacturers in 1997 to address the losses

Leading global association for building resilience and sustainability using its library of standards and best practices.

Unites global manufacturers, logistics providers, freight carriers, law enforcement agencies, and other stakeholders.

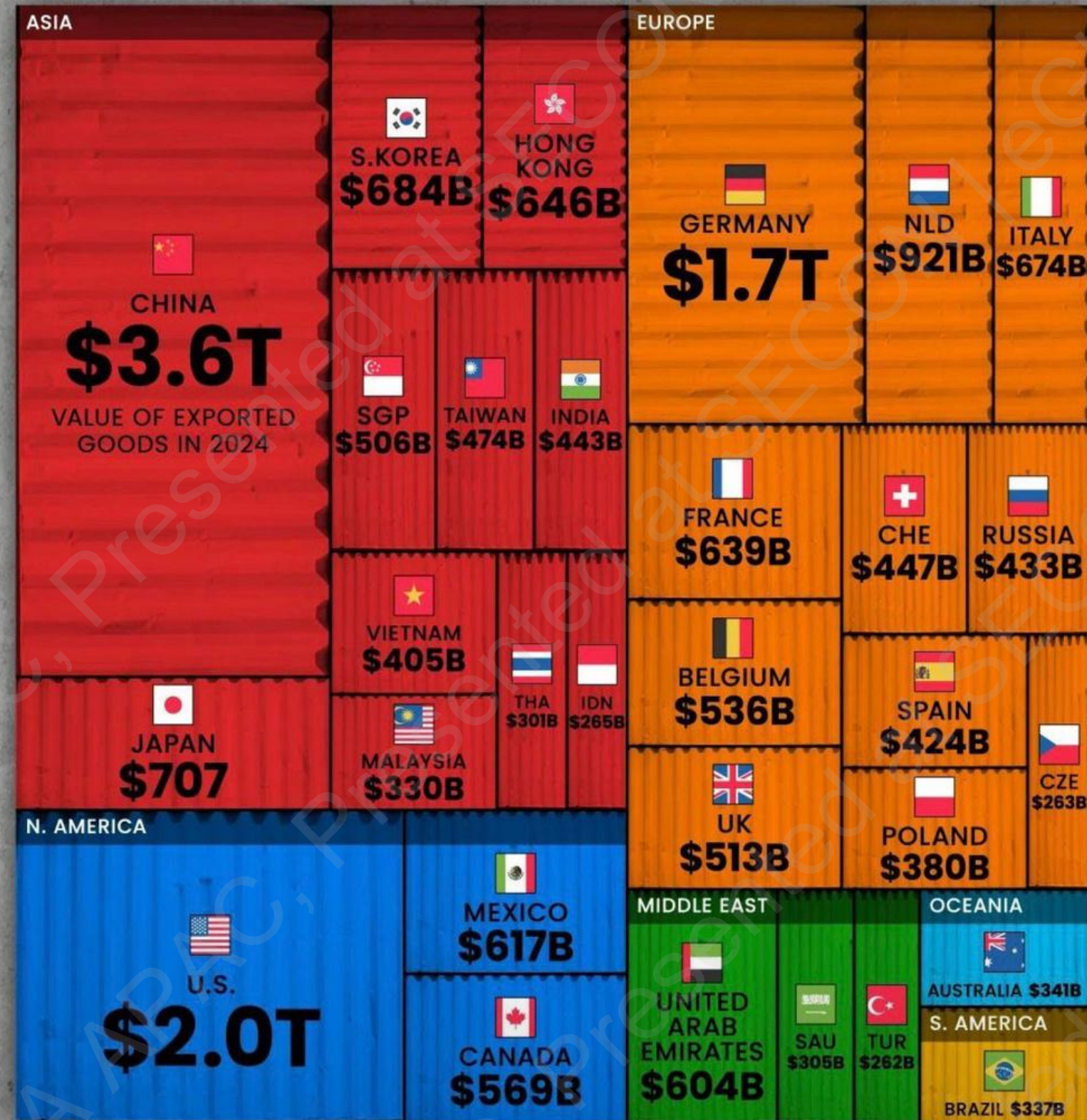
The core concept of TAPA is to mitigate losses through the implementation and enforcement of defined contractual supply chain security and resilience requirements.



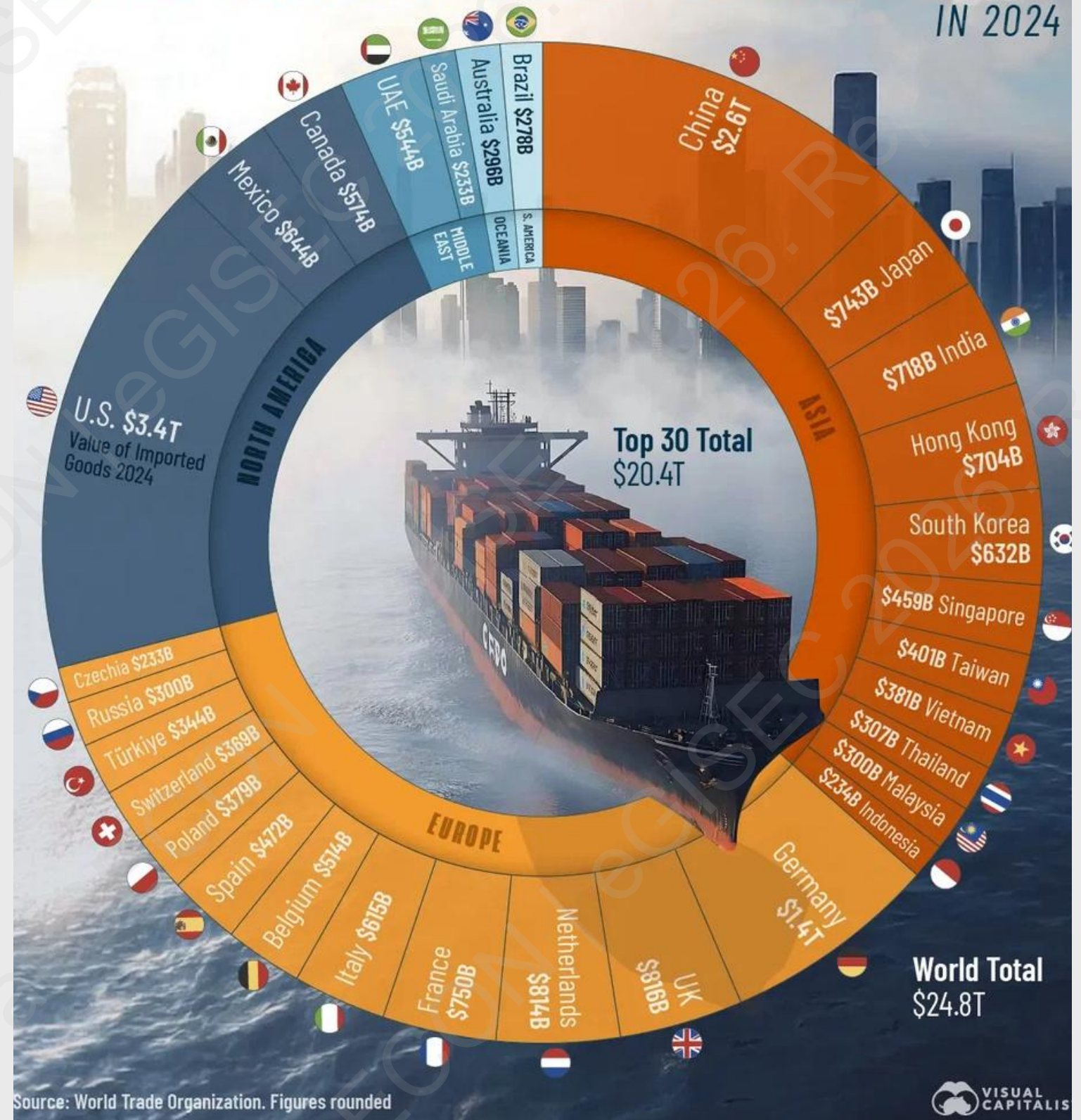
THE WORLD'S Biggest Exporters

TOP 30 TOTAL **\$20.3T**

WORLD TOTAL **\$24.5T**



THE WORLD'S 30 BIGGEST IMPORTERS IN 2024



Source: World Trade Organization. Figures rounded

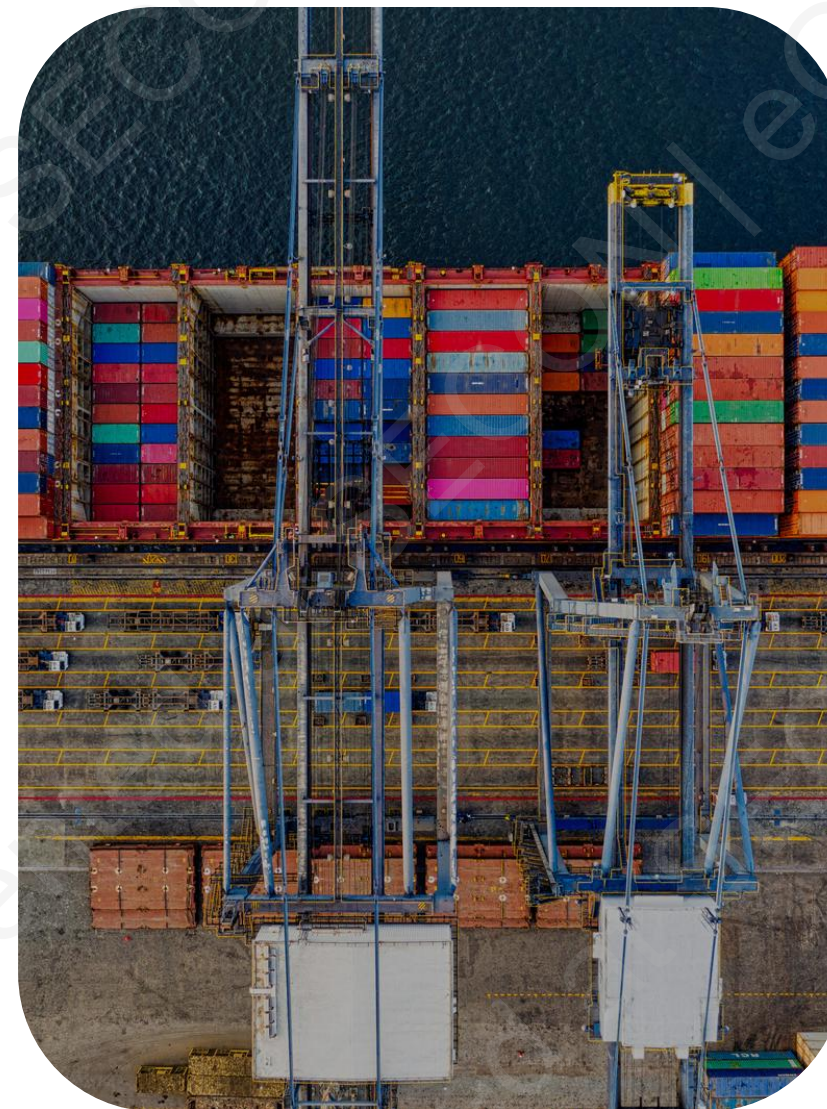
VISUAL CAPITALIS

The Challenge

A Volatile and Fragmented World



**Multipolar Trade and
Tariff Tensions**



**Rising Cargo Crime,
Counterfeiting, and
Digital Fraud**



**Data Silos and
Fragmented Standards**



**Increasing ESG and
Compliance Demands**

Case Studies

The threat to our increasingly complex web of global supply is real. But organizations must look beyond their software to successfully mitigate supply chain disruption. Securing hardware and logistics is crucial, as these supply chain links are also in the crosshairs of nation states.

In July 2024, suspicious packages caused a warehouse fire at a depot belonging to courier giant in Birmingham, UK. British counter-terrorism authorities investigated Russian involvement in this fire, which was linked to a very similar incident that same month at another facility in Germany.

In September 2024, camouflaged explosives were hidden in walkie talkies and pagers in Lebanon and Syria – a supply chain attack widely believed to be carried out by Israel.

While these attacks targeted hardware and logistics rather than software, the underlying rule of thumb remained the same: the compromise of a single distributor can provide the attackers with considerable economies of scale.

Question: how do we address this?

FSR

IMS

RAS

CSS

CAPTURED INCIDENTS OF 2023 (GLOBAL)

APAC INCIDENT DATA FOR 2023

COMBINED LOSS IN 2023

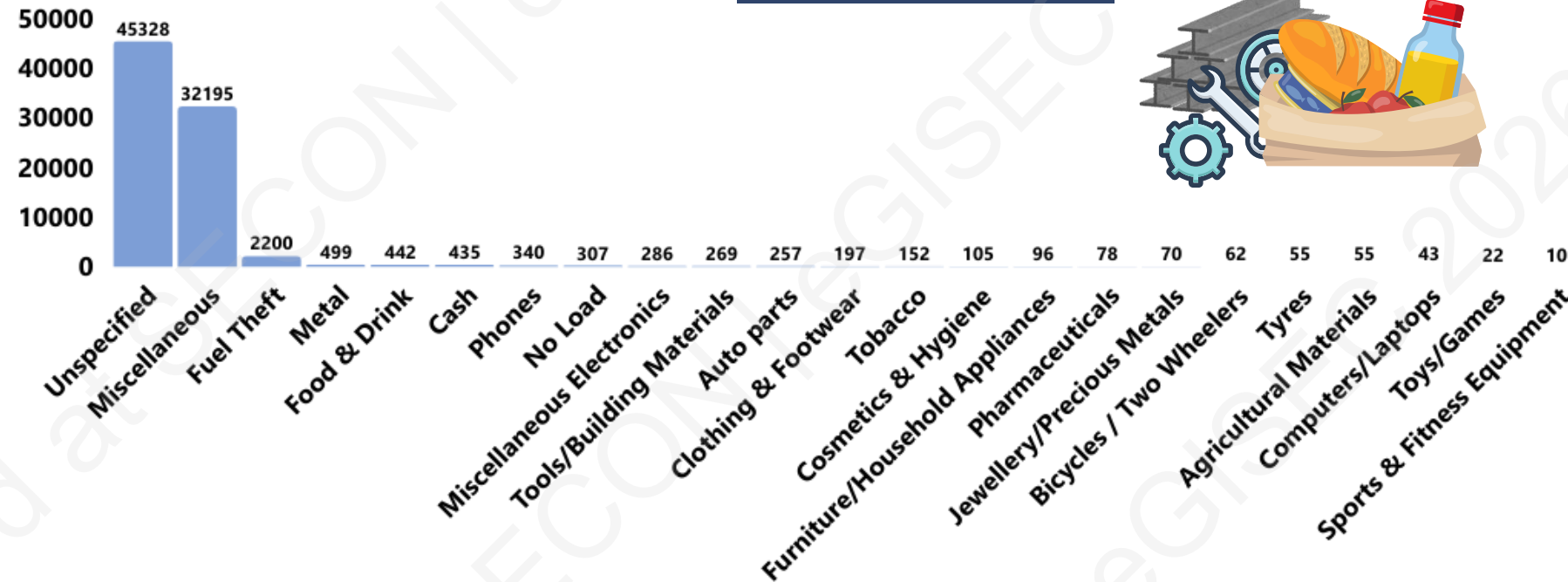
USD
\$931,908,997

HIGHEST CASES IN 2023

BY COUNTRY

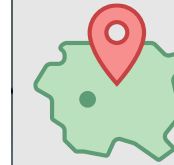
Countries	No. of Incidents
South Africa	69,108
United Kingdom	4,211
Germany	2,816
France	1,289
Italy	1,117
Sweden	1,091
Spain	965
Poland	216
Netherlands	215
India	201

INCIDENTS REPORTED BY PRODUCT CATEGORY



Total Recorded Cargo Crime Incidents Globally in 2023

83,484



Countries Reported Incidents in 2023

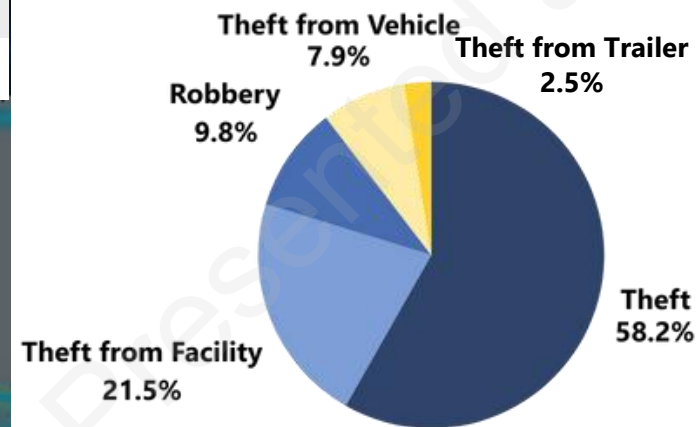
133



Cases globally reporting a loss value

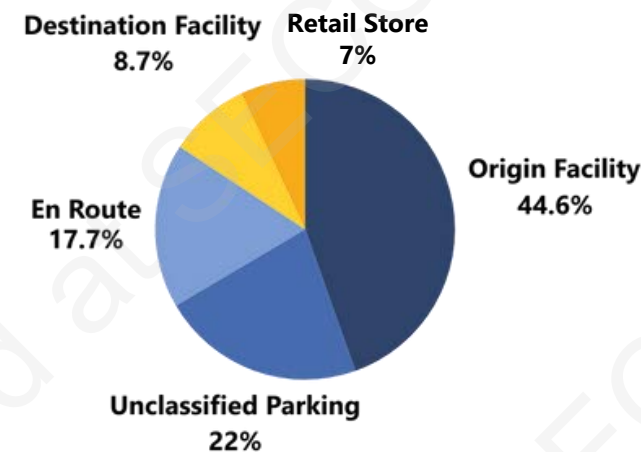
4,177

TOP 5 INCIDENTS REPORTED BY INCIDENT CATEGORY



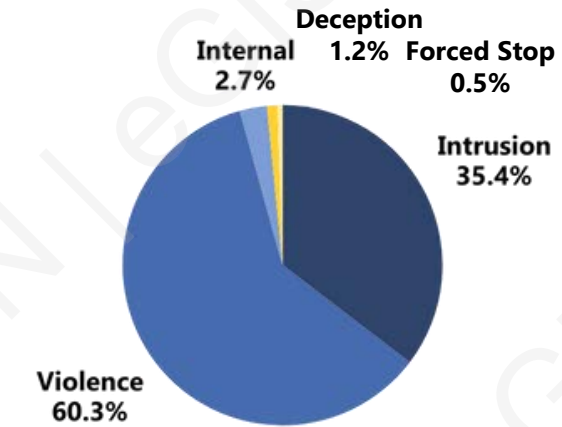
Theft
58.2%

TOP 5 INCIDENTS REPORTED BY LOCATION TYPE



Origin Facility
44.6%

TOP 5 INCIDENTS REPORTED BY MODUS OPERANDI



Violence
60.2%

For more information about TIS, please email to us at tis@tapa-apac.org

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APAC INCIDENT DATA FOR 2024

COMBINED LOSS IN 2024

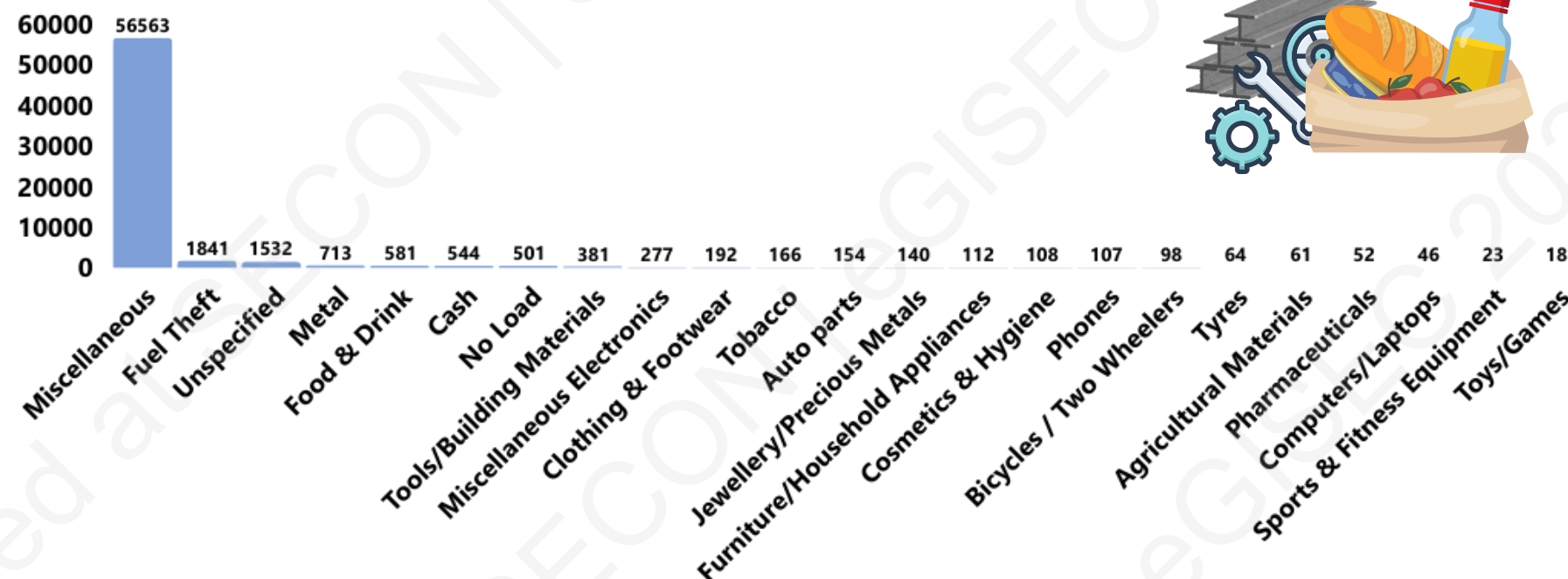
USD
\$2,203,247,619

HIGHEST CASES IN 2024

BY COUNTRY

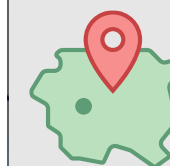
Countries	No. of Incidents
South Africa	47,385
Germany	3,506
United Kingdom	2,975
Italy	2,020
France	1,707
Spain	1,355
Sweden	631
Brazil	500
Netherlands	436
India	316

INCIDENTS REPORTED BY PRODUCT CATEGORY



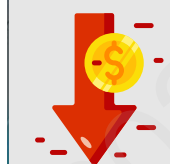
Total Recorded Cargo Crime Incidents Globally in 2024

64,274



Countries Reported Incidents in 2024

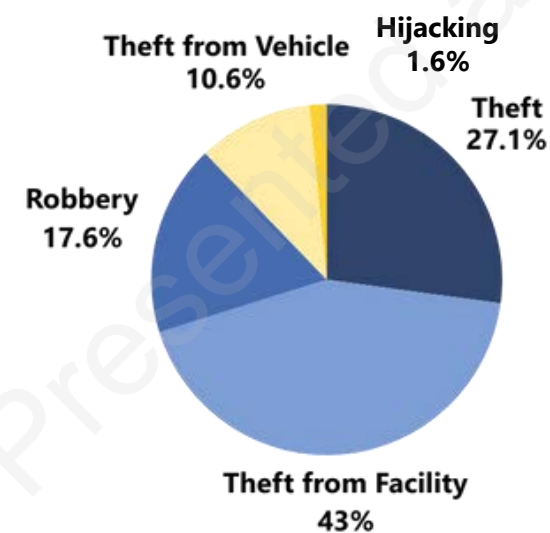
147



Cases globally reporting a loss value

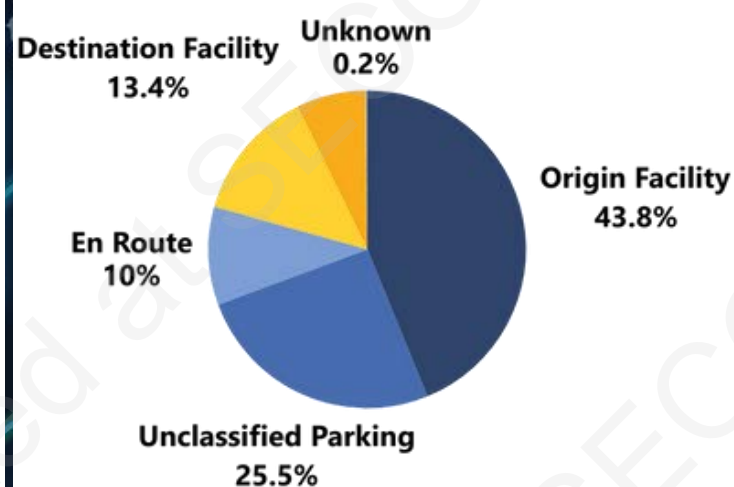
3,511

TOP 5 INCIDENTS REPORTED BY INCIDENT CATEGORY



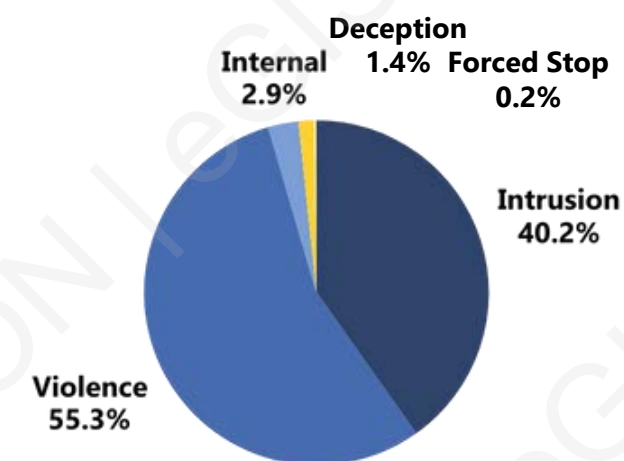
Theft from Facility
43%

TOP 5 INCIDENTS REPORTED BY LOCATION TYPE



Origin Facility
43.8%

TOP 5 INCIDENTS REPORTED BY MODUS OPERANDI



Violence
55.3%

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APAC INCIDENT DATA FOR 2025

DIRECT LOSS IN 2025

USD **\$686,083,462**

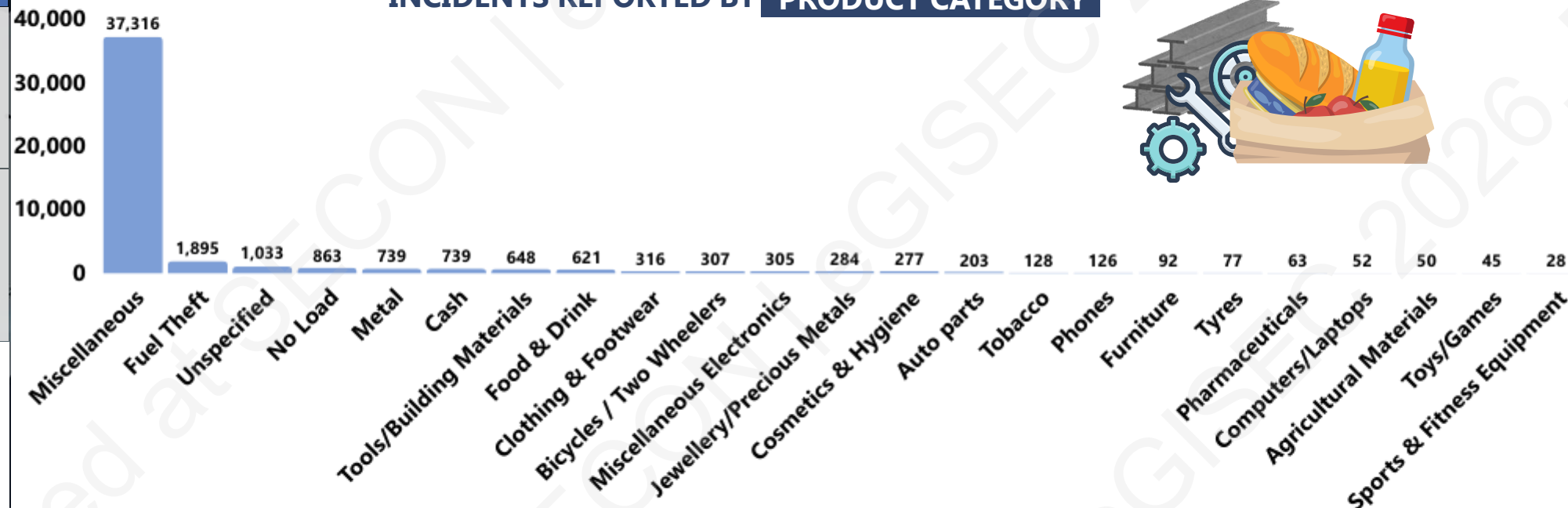
INDIRECT LOSS IN 2025

USD **\$2,463,930,820**

HIGHEST CASES IN 2025 BY COUNTRY

Countries	No. of Incidents
South Africa	22451
Germany	4836
Italy	3550
United Kingdom	2658
France	2489
Spain	2334
Sweden	1561
Poland	1230
Czechia	690
Netherlands	655
Slovakia	341
Belgium	282
India	236
Brazil	218
Greece	216
Romania	201
Russian Federation	187
Switzerland	186

INCIDENTS REPORTED BY PRODUCT CATEGORY

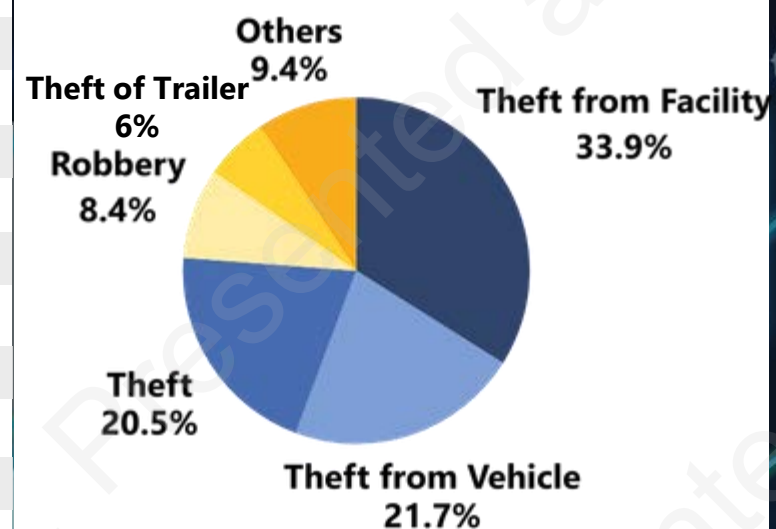


Total Recorded Cargo Crime Incidents Globally in 2025 **46,205**

Countries Reported Incidents in 2025 **117**

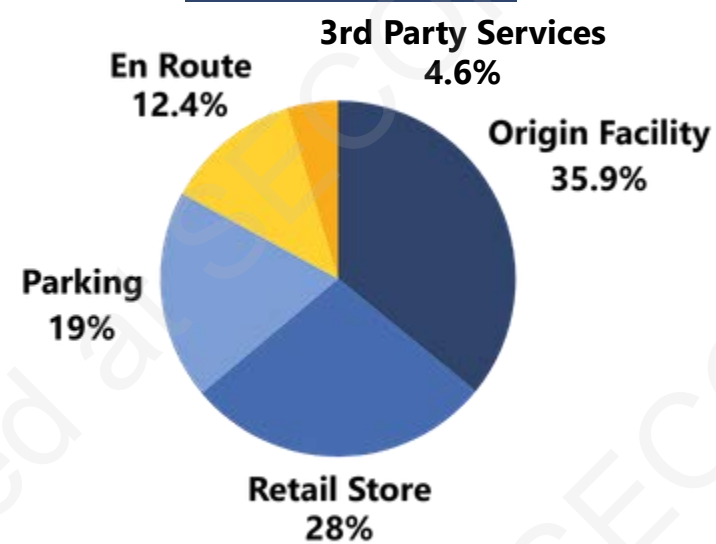
Cases globally reporting a loss value **3,198**

TOP 5 INCIDENTS REPORTED BY INCIDENT CATEGORY



Theft from Facility
33.9%

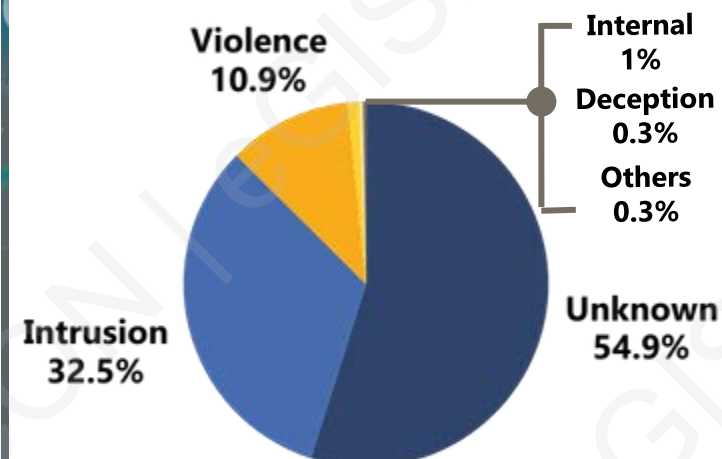
TOP 5 INCIDENTS REPORTED BY LOCATION TYPE



Origin Facility
35.9%

*Excludes the "Unknown" category

TOP 5 INCIDENTS REPORTED BY MODUS OPERANDI



Intrusion
32.5%

Counterfeit Cases **605**

Counterfeit cases that stated a value **303**

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TIS CAPTURED INCIDENTS OF 2025 (APAC)

APAC INCIDENT DATA FOR 2025

DIRECT LOSS IN 2025

USD
\$30,466,734

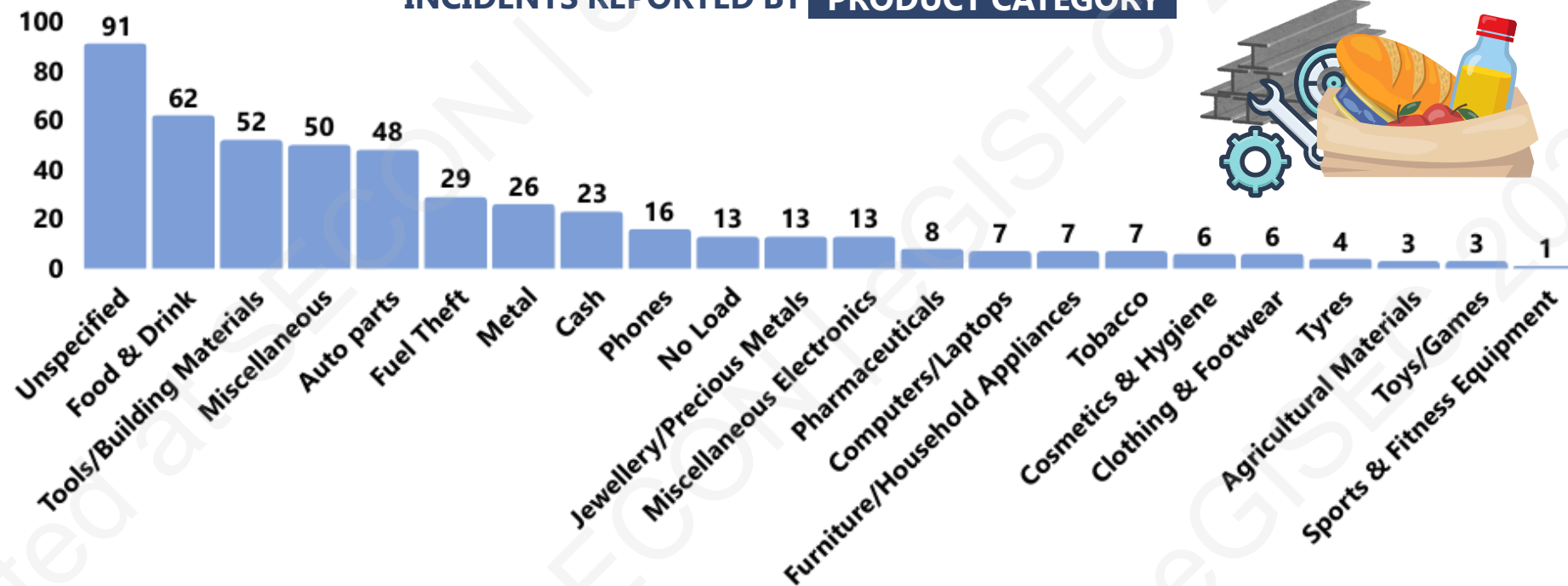
INDIRECT LOSS IN 2025

USD
\$2,409,756,398

CARGO THEFT IN 2025 BY COUNTRY

Countries	No. of Incidents
India	235
Indonesia	120
Malaysia	21
Australia	18
Philippines	16
Pakistan	15
Singapore	14
Thailand	13
Hong Kong	8
Bangladesh	6
South Korea	5
Viet Nam	4
Macao	3
New Zealand	3
Japan	3
China	2
North Korea	1
Myanmar	1

INCIDENTS REPORTED BY PRODUCT CATEGORY



Total Recorded APAC Cargo
Crime Incidents in 2025

488



APAC Countries Reported
Incidents in 2025

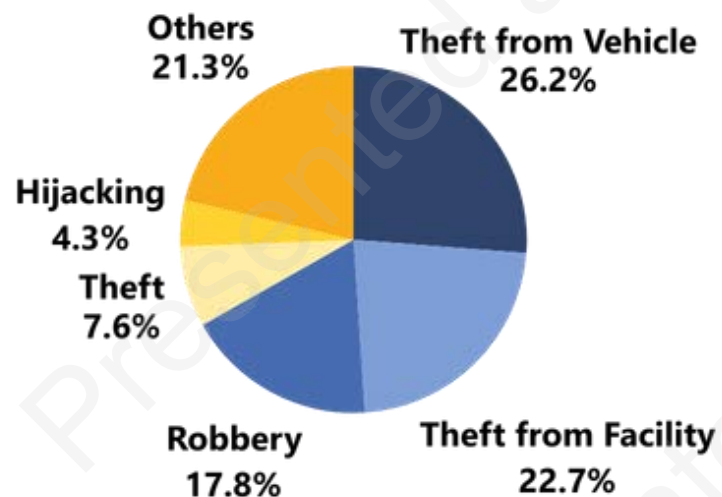
18



Cases in APAC
reporting a loss value

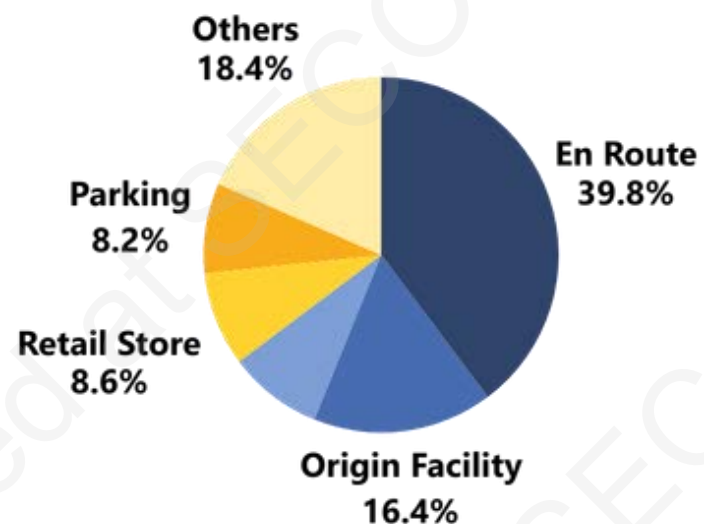
196

TOP 5 INCIDENTS REPORTED BY INCIDENT CATEGORY



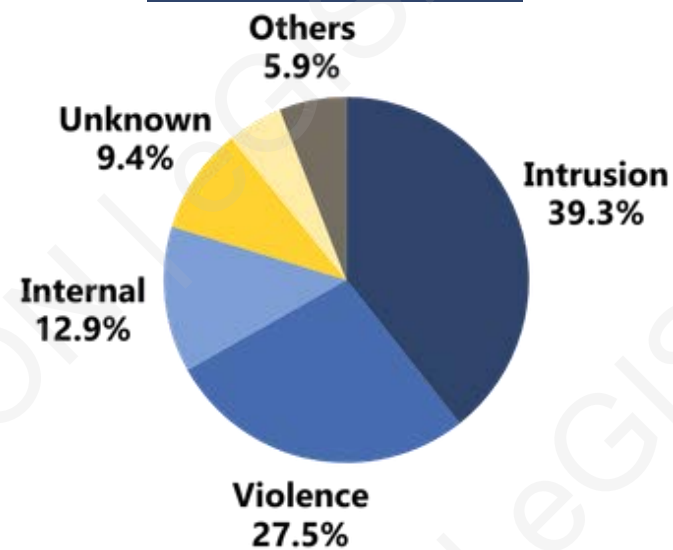
Theft from Vehicle
26.3%

TOP 5 INCIDENTS REPORTED BY LOCATION TYPE



En Route
39.8%

TOP 5 INCIDENTS REPORTED BY MODUS OPERANDI



Intrusion
39.2%

Counterfeit Cases

516

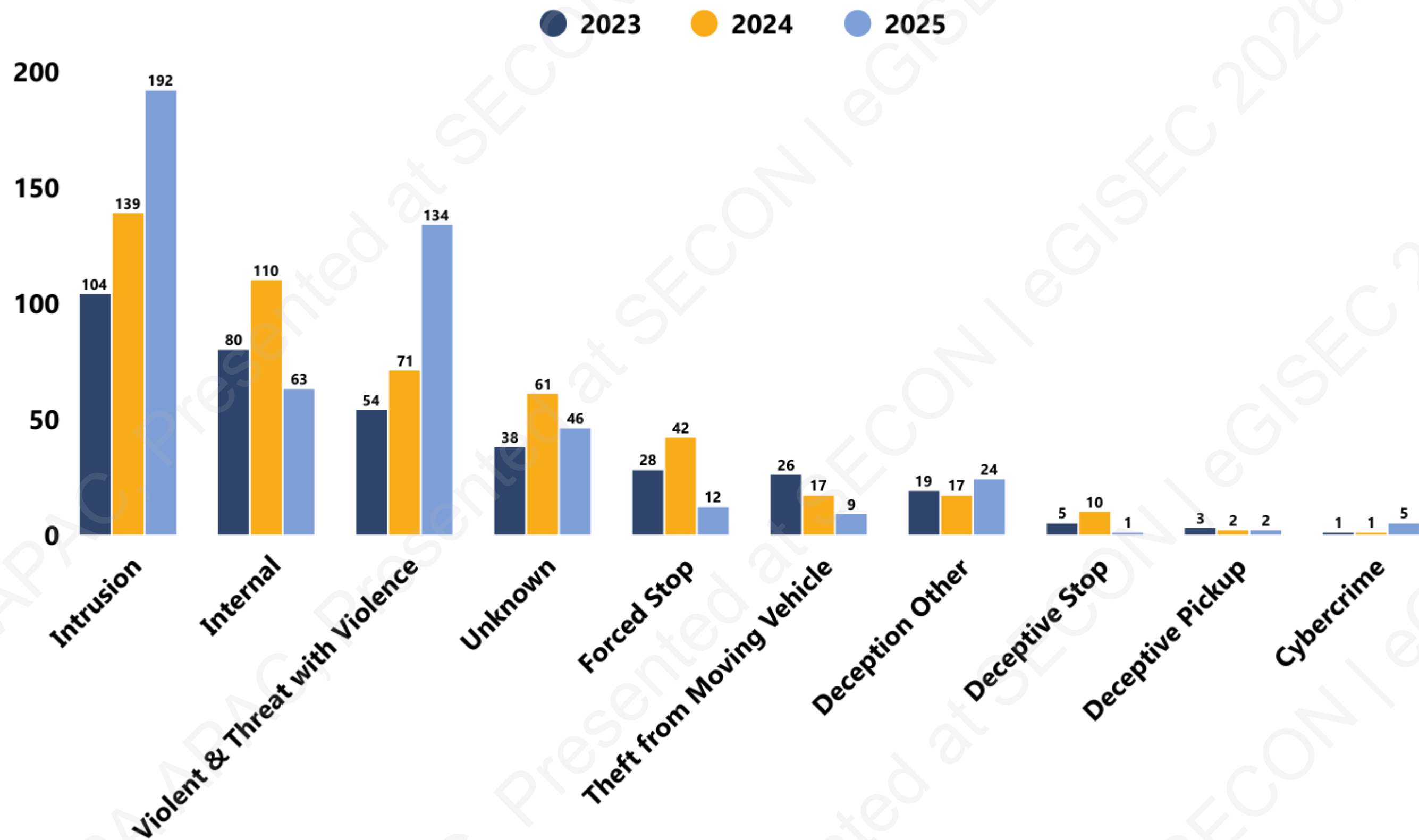
Counterfeit cases that
stated a value

295

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Modus Operandi Year-on-Year Comparison



Modus Operandi	Y23	Y24	Y25
Intrusion	104	139	192
Internal	80	110	63
Violent & threat with Violence	54	71	134
Unknown	38	61	46
Forced Stop	28	42	12
Theft from Moving Vehicle	26	17	9
Deception Other	19	17	24
Deception Stop	5	10	1
Deception Pickup	3	2	2
Cybercrime	1	1	5

TAPA APAC Intelligence System (TIS)

◦ Global database of cargo crime including counterfeit incidents, and TAPA-certified sites

- Total GLOBAL incidents: 266,849
- Total APAC incidents: 4,423
- Total GLOBAL pharma incidents: 881
- Total APAC pharma incidents: 133
- Total GLOBAL counterfeit incidents: 902
- Total APAC counterfeit incidents: 720

◦ Anonymous reporting

◦ Smart cargo route planning

◦ Data extraction and reporting across regions



TAPA APAC INTELLIGENCE SYSTEM

COUNTERFEIT INCIDENT ON 24.07.2024

Incident Details

DATE
24.07.2024

INCIDENT CATEGORY
Counterfeit

MODUS OPERANDI
Deception Other

Place

CITY
-

REGION / COUNTRY
- / Hong Kong

LOCATION TYPE
En Route

Product

Product category percentages for incidents reported



Product category	Incidents
Food & Drink	90 24.7%
Miscellaneous	36 9.9%
Tools/Building Materials	29 7.9%
Metal	29 7.9%
Auto parts	26 71%



MEDIA REPORT

three shipping

<https://www.scmp.com/news/hong-kong/law-and-crime/article/3271895/customs-officers-seize-hk140-million-black-market-cigarettes-after-sea-chase>

Export incident data

DOWNLOAD PDF

DOWNLOAD CSV



TAPA Standards



Facility Security Requirements (FSR)

Warehousing & In-Transit Storage



Trucking Security Requirements (TSR)

Trucking & Road Transportation



Guarding Security Requirements (GSR)

Guarding Operations



Risk Assessments Standard (RAS)

End-to-End Supply Chain Risk Identification, Analysis, Evaluation & Mitigation



Cyber Security Standard (CSS)

Protection Against Cyber Threats



Pharmaceutical Facility Security Requirements (PFSR)

Pharmaceutical Facility Storage



Business Continuity Planning (BCP)

Safeguarding Business Operation



Air Cargo Security Standard (TACSS)

Air Transportation

Setting the Stage & Opportunity



- Fragile systems break under stress.
- Resilient systems resist shock.
- Anti-fragile systems grow stronger from disruption.
- The new supply chain paradigm: Predict. Adapt. Strengthen.
- Leverage AI, IoT, and blockchain for predictive intelligence and traceability.
- Embed TAPA Standards (FSR, TSR, BPS, Cybersecurity) as digital trust anchors.
- Build a collaborative ecosystem across shippers, LSPs, insurers, and regulators.

Visibility Through IoT

IoT: The Nervous System of the Cargo Network

- Smart sensors & GPS tracking create real-time situational awareness.
- Predictive maintenance & route optimization reduce risk.
- Integration with TAPA Trucking Security Requirements (TSR) ensures secure, compliant transport monitoring.



Intelligence Through AI

AI: From Data to Foresight

- Detect anomalies and predict theft or route deviations.
- Use TAPA Intelligence System (TIS) for proactive threat analytics.
- AI-enabled audit & compliance tools ensure adaptive governance.



Trust Through Blockchain

Blockchain: Securing the Chain of Custody

- Immutable digital ledgers for cargo movement and handover.
- Strengthen traceability and authenticity for high-value goods.
- Smart contracts automate compliance with TAPA FSR/TSR/BPS standards.



Digital Compliance in Action

From Manual Checklists to Smart Frameworks

- Integrate TAPA standards into digital twins and audit dashboards.
- AI maps compliance gaps and suggests corrective actions.
- Transparency for regulators, insurers, and brand owners.



Human + Machine Collaboration

Technology Amplifies — People Empower

- Cross-functional collaboration between logistics, IT, and security teams.
- Shared intelligence within TAPA member networks.
- Continuous learning through TAPA Training Academy and LMS.



Case Study

Secure by Design: The Anti-Fragile Model

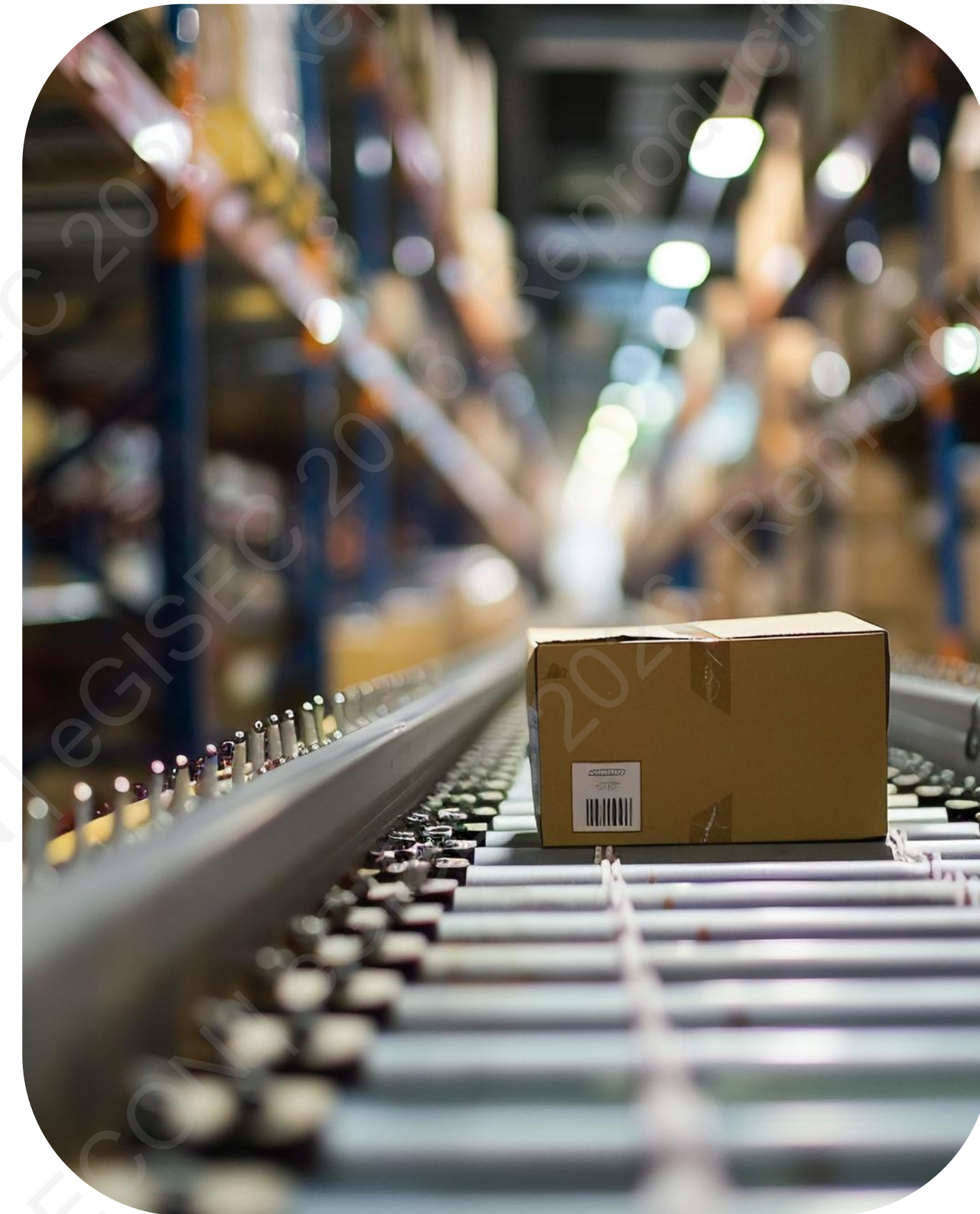
- Example: IoT-enabled cold chain with blockchain audit trail.
- AI alerts integrated with TAPA TSR & FSR certification dashboards.
- Results: 40% reduction in theft risk, 25% faster incident response.



The Future Outlook

Towards a Digital Trust Network

- Convergence of physical and cyber security standards.
- AI-driven certification and risk scoring.
- TAPA's evolving ecosystem: 2026 Standards, digital audits, and collaboration.





Global Standards

- Structured framework with global recognition
- Industry best practices
- Scalable and flexibility
- Alignment between stakeholders
- Common minimum standard written in contractual agreements

Intelligence



- Cargo crime and counterfeit incident data
- TAPA-certified locations
- TIS e-Alert
- Quarterly and annual reports with insights

Training, Audit & Certifications

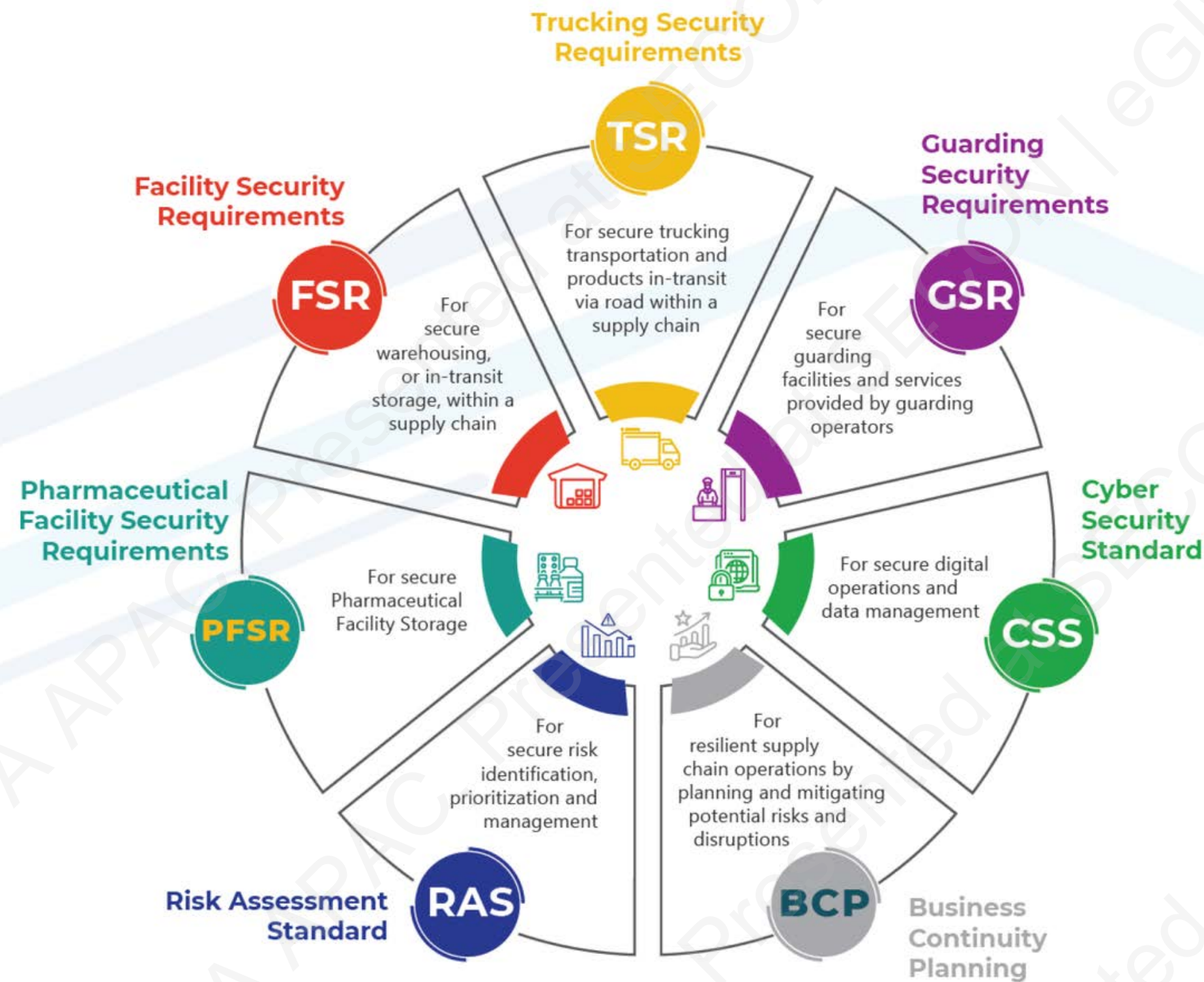
- Industry expert trainers
- Self-paced Learning Management System (LMS)
- Blockchain certification
- Audit partners and self-audit
- Multisite certifications

Collaborative Community

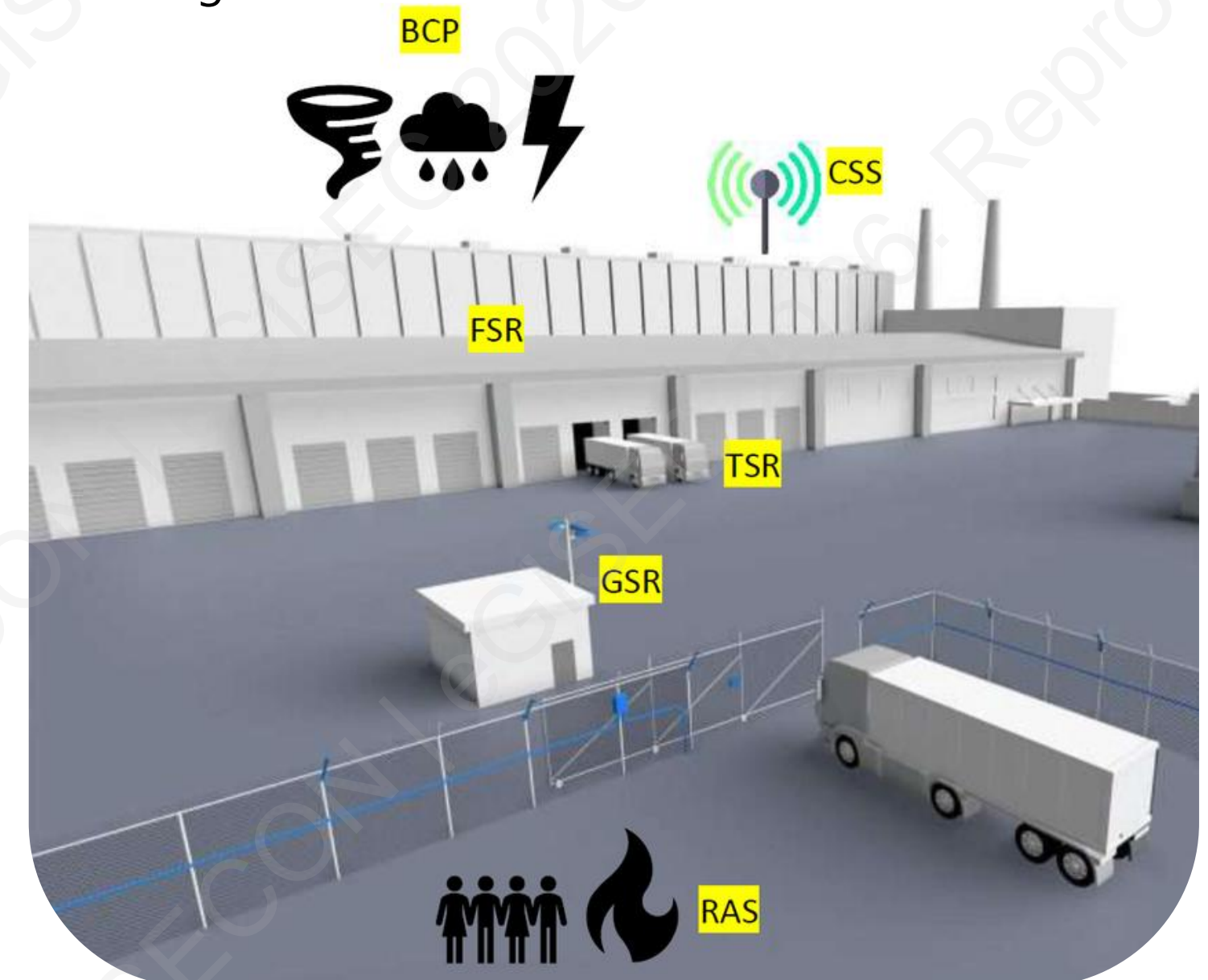
- TAPA Partnerships with governments and Law Enforcement Agencies (LEAs)
- Standards working groups
- Networking events, SCRS Conference, seminars, webinars

TAPA Standards

An Ecosystem of Resilient Supply Chain Standards



The Transported Asset Protection Association Asia Pacific (TAPA APAC) has developed a suite of globally recognized security standards to safeguard critical assets throughout the supply chain. Adopting these standards delivers both operational efficiency and risk mitigation.



TAPA APAC Community!

Some of Our
850+ Member
Companies from
Global
Superbrands..

For the full list of TAPA APAC member companies, please visit:
<https://tapa-apac.org/member-companies/>

TAPA APAC
Transported Asset Protection Association



And many more...

Join us at our upcoming events!

NGO WEEK EVENT

JUNE-JULY 2026
CHINA, SHANGHAI

CHINA

PUDONG SHIPPING WEEK

DECEMBER 2026
CHINA, SHANGHAI



VIETNAM SUPERPORT™

2 JUNE 2026
NEAR HANOI, PHU THO
PROVINCE, VIETNAM,

VIETNAM



SCRS SUPPLY CHAIN RESILIENCE & SUSTAINABILITY

CONFERENCE & AWARDS 2026

15 & 16 SEPTEMBER 2026
SINGAPORE

SINGAPORE



SCAN THE QR CODE





The Leading Association for Global Supply Chain Resilience & Sustainability

For more information, please contact TAPA APAC at info@tapa-apac.org

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